Basic ideas on Singular Value Decomposition (SVD)

Numerical Linear Algebra

The SVD decomposition

We consider $A \in \mathbb{R}^{m\times n}$. A can be written as

$$A = USV^t \tag{1}$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal, $S \in \mathbb{R}^{m\times n}$ is diagonal and

- S contains contains the singular values (singular values $= \sqrt{|\text{eigenvalues of } AA^t \text{ or } A^tA|}$). The singular values appear ordered arranged in descending order.
- The columns of U are left unitary singular vectors (i.e. the eigenvectors of AA^t are the columns of U).
- The rows of V^t are right unitary singular vectors (i.e. the eigenvectors of A^tA are the columns of V).

Remarks:

1. The SVD represents an expansion of the original data in a coordinate system where the covariance matrix is diagonal (it gives information on the principal components (PCA)).
2. The relation $AV = US$ means that there exists a special orthonormal set of vectors (i.e. the columns of V), that is mapped by the matrix A into an orthonormal set of vectors (i.e. the columns of U).
3. The singular values are always real numbers (because A^tA and/or AA^t are symmetric matrices).
4. Recall that eigenvectors of different eigenvalues of symmetric matrices are orthogonal, hence U and V are orthogonal matrices. This follows from the fact that

   $$\langle Ax, y\rangle = \langle x, A^ty\rangle$$

   Then, if x, y are eigenvectors of different eigenvalues $\lambda \neq \mu \neq 0$ of A it follows that $\lambda\langle x, y\rangle = \langle Ax, y\rangle = \langle x, A^ty\rangle$ and if A symmetric $\langle x, A^ty\rangle = \mu\langle x, y\rangle$ and hence $\langle x, y\rangle = 0$.
5. The SVD decomposition is not unique, it is only unique up to a reflection of each of the singular vectors (because $s_ku_kv_k = s_k(-u_k)(-v_k)$).

Ex.1 Code a simple algorithm to compute SVD decomposition of a matrix A using the eigenvalues/eigenvectors of A^tA and AA^t.

Ex.2 Use the `scipy.linalg.svd` function to get the SVD decomposition of A.

Use of SVD for some linear algebra computations

We recall that:

1. The 2-norm of A is the maximum of the spectra radius of A^tA,
2. The Frobenius norm of A is equal to the square root of the sum by rows and columns of the square of the elements of A.
3. The (Moore-Penrose) pseudoinverse of $A = USV^t$ (thin SVD) is the matrix $A^+ = VS^{-1}U^t$ (for a diagonal matrix one has $S^+ = S^{-1}$).

Ex.3 Write a program that uses SVD decomposition to compute

(a) the rank(A),
(b) the 2-norm of A,
(c) the Frobenius norm of A,
(d) the condition number $k_2(A)$,
(e) the pseudoinverse A^+ of A.

Computing the SVD decomposition

There are different strategies to compute the SVD of a matrix $A \in \mathbb{R}^{m\times n}$. Below we describe a method to compute the singular values.

The method consists in two main steps:

Step 1. Bidiagonalize the matrix

$$H = \begin{pmatrix} 0 & A^t \\ A & 0 \end{pmatrix}$$

to obtain $B \in \mathbb{R}^{(m+n)\times(m+n)}$ upper bidiagonal.

Step 2. Use LR-type iteration to diagonalize B and obtain the singular values in the diagonal.

Ex.1 Write a routine to check that:

(a) the eigenvalues of H are $\pm s_i$, $i = 1 \dots n$, where s_i, $i = 1, \dots, n$ are the singular values of A.

(b) if v is an eigenvector of H then $\sqrt{2}v$ is a column of

$$\begin{pmatrix} V \\ \pm U \end{pmatrix} \text{ or } \begin{pmatrix} -V \\ \pm U \end{pmatrix}$$

The bidiagonalization process can be carried out applying Householder transformations.

Ex.2 Write a function `house(x)` such that given $x \in \mathbb{R}^l$ computes $v \in \mathbb{R}^l$, with $v_1 = 1$, and $\beta \in \mathbb{R}$ so that the Householder transformation $P = I - \beta vv^t$ is such that $P(x) = \|x\|_2 e_1$, $e_1 = (1, 0, \dots, 0)$. This can be achieved by the algorithm 5.1.1. in Matrix computations, Golub-Van Loan, 3rd ed. p210.

In practice, one never forms the matrix P explicitly. Note that one has

$$PA = (I - \beta vv^t)A = A - vw_1^t$$

where $w_1 = \beta A^t v$ and

$$AP = A(I - \beta vv^t)A = A - w_2 v^t$$

where $w_2 = \beta Av$.

Ex.3 Write functions `PA(bet,v,A)` and `AP(bet,v,A)` that perform the previous updating computations.

Ex.4 Write a function `bidiag(A)` that performs the bidiagonalization of A by applying Householder transformations. If $A \in \mathbb{R}^{m\times n}$, one step of the algorithm consist in

(a) remove the terms below the diagonal (in the column j) using a Householder transformation P,

(b) update $A = PA$,

(c) remove the terms to the right of the superdiagonal (of the row j),

(d) update $A = AP$.

This is performed for $j = 1, \ldots, n$ steps.

Ex.5 Write a program that giving a matrix A, computes the matrix H and reduce it to bidiagonal B. Write the output to a file (just the dimension $m+n$ and the two arrays containing the bidiagonal of H).

This bidiagonal matrix B will be the input of the LR-algorithm.

Ex.6 Write a program that implements the *qds* algorithm 5.10 in Applied Numerical Linear Algebra, Demmel, p244. The algorithm converges to a diagonal matrix with the square of the singular values of B in the diagonal. Let a and b are the diagonal and superdiagonal, respectively, of $B \in \mathbb{R}^{n\times n}$. Then consider $q = a^2$ and $e = b^2$. One step of the algorithm is given by

$$\begin{aligned} &for\ j = 1\ to\ n-1 \\ &\quad \hat{q}_j = q_j + e_j - \hat{e}_{j-1} \\ &\quad \hat{e}_j = e_j(q_{j+1}/\hat{q}_j) \\ &end\ for \\ &\hat{q}_n = q_n - \hat{e}_{n-1} \end{aligned}$$

where we assume that $b_0 = \hat{b}_0 = b_n = \hat{b}_n = 0$. Stop the iteration whenever $\|e\|_\infty < 10^{-14}$.